ST. GODFREY (P.V.T.) SECONDARY SCHOOL

2023 MSCE END OF TERM ONE EXAMINATION

CHEMISTRY

Theory

Instructions:

1. **This paper contains 9 printed pages. Please check.**
2. There are **two** sections in this paper; **A** and **B**.
3. Answer **all** questions in **all** sections.
4. Write your **Name** on top of each page of your question paper.
5. In the table provided on this page, **tick** against the question number you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			

Section A (70 marks) Answer **all** questions from this section.

1. Sodium (Na) reacts with water (H_2O) to produce sodium hydroxide (NaOH) and hydrogen gas (H_2) according to the following chemical equation

$$Na_{(s)} + H_2O_{(l)} \longrightarrow NaOH_{(aq)} + H_{2\,(g)}$$

a. Balance the equation.

_____

_____

(3marks)

b. If 46g of sodium metal reacts completely with 18g of water, what mass of sodium hydroxide is produced?

(2marks)

c. What volume of hydrogen gas is produced in the reaction 1 (a) above assuming the reaction takes place at room temperature and pressure?

(2marks)

d. (i) Explain how surface area of reactants affect the rate of chemical reaction.

_____

_____

_____

_____

(2marks)

(ii) Why catalyst do not affect the position of the equilibrium in a reversible chemical reaction?

_____

_____

_____

(1mark)

2.a. Define an acid according to Lowry-Bronsted theory.

_____

_____

(1mark) b. Outline any **two** properties of a base.

_____

_____

(2 marks)

c. Complete the following by writing the conjugate acid-base (4marks)

i. H_2SO_4 + NH_3 _____ + _____ ii.

HNO_3 + H_2O _____ + _____

(3 marks)

d.Explain how hydronium ion is produced.

_____

_____

_____

(2marks)

e. Explain how you can test the strength of an acids.

_____

_____

_____

_____

(2marks)

3.a. Define Preferential discharge.

_____

_____

(1mark)

b.Mention **three** factors affecting preferential discharge.

_____

_____

_____

(3 marks) c. Give **three** uses of electrolysis.

_____

_____

_____

(3 marks)

d. Give the name of the precipitate formed when Lead Nitrate solution is mixed with Sodium bromide.

_____

(1 mark)

4. a. Mention any **two** allotropes phosphorus.

_____

_____

(2marks) b. Give any **two** uses of sulphates.

_____

_____

(2 marks)

5. a. (i) Which element is more reactive between Sodium (Na) and Magnessium (Mg)?

_____

(1mark) (ii) Give reason for your answer to question 5. a. (i).

_____

_____

(2marks)

b. Gabriellium is a newly discovered metal element by form 4 learners at Namtegha CDSS. After taking a series of tests, they found that the element is very soft and reacts vigorously with water giving out a gas that produces a pop sound to a glowing splint. Its solution turns red litmus paper to blue.

(i) Suggest the family of this element.

_____

(1mark)

(ii) Would this element be found along the banks of river?

_____

(1mark) (iii) Suggest a reason for your answer to question 5 d(ii).

_____

_____

(1 mark) (iv) Suggest how you can keep Gabriellium in the laboratory.

_____

(1mark) (v) Give the name of the gas that was produced from the reaction.

_____

(1mark)

6. a. Define waste management.

_____

_____

(1mark)

b. Give any **three**major sources of wastes.

_____

_____

_____

(3 marks)

c. Explain how compost manure is one way of solid waste disposal.

_____

_____

_____

d. Give any **two** benefits of recycling waste.

_____

_____

_____

_____

(2marks)

7. a (i)Give the importance of ozone layer.

_____

_____

(1mark)

(ii) Mention any **two** ways of protecting the ozone layer.

_____

_____

_____

_____

(2marks) (iii) Explain how combustion is a source of greenhouse gases.

_____

_____

(1mark)

b. Write any **two** effects of global warming.

_____

_____

_____

_____

(2marks)

8. a. Differentiate between hard water and soft water.

_____

_____

(1 mark)

b. Give **two** methods of removing hardness in water.

_____

_____

(2marks)

c. Mention **two** importance of hard water to everyday life.

_____

_____

(2marks)

d. Calculate the concentration in moles per litre of 40% Sodium hydroxide solution (40% NaOH). RAM, Na = 23, O = 16 and H = 1.

(3 marks)

9. a. What is oxidation in terms of redox reaction?

_____

(1mark)

b. Study the chemical equation below and use it to answer questions that follow:

$2FeCl_{2\,(aq)} + Cl_{2\,(g)} \rightarrow 2FeCl_{3\,(aq)}$

i. Write the net equation for the reaction.

_____

(1mark) ii. Write oxidation half equation for the reaction.

_____

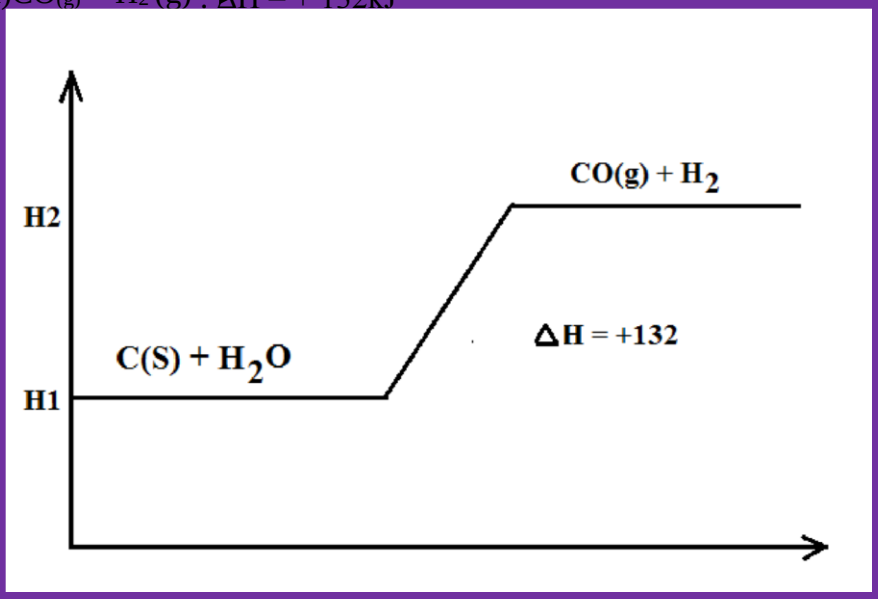
(1mark) iii. Write reduction half equation for the reaction.

_____

(1mark)

10. Draw the heat energy level diagram for the reaction below:

$C_{(s)}$ + $H_2O(l)CO_{(g)} + H_2(g)$; $\Delta H = +132kJ$



(3 marks)

Section B (30 Marks)

11. a. Differentiate between addition and condensation polymerization.

_____

_____

_____

(2marks)

b.Using the structures of monomer, show how polyvinylchloride (PVC) is formed.

(2marks)

c. With a well labelled diagram explain how ethanol is produced by fermentation of sugar.

(6marks)

12. Study the chemical equations bellow and use them to answer the questions that follow:

Fe + Mg^{+2}	No reaction
Mg^{+2} + Al	No Reaction
Al^{+3}+ Fe	No Reaction
Cu^{2+}+ Pb	Reaction
$Pb2^{+}$+ Fe	Reaction

a. Arrange the elements in order of increasing reactivity.

_____

_____

_____

_____

_____

(5marks)

b.Write down the products formed when aluminum metal is dipped in a solution of lead._____

_____

____

(2marks)

c. Draw an electrochemical cell for the reaction in 12 (**b**) above, label any **three** parts.

(3marks)

13. a. With the help of the diagram, explain how ammonia is produced in Haber process.

_____

_____

_____

_____

(7marks)

b. Draw the dot cross diagram for the formation of Aluminum trioxide (Al_2O_3).

(3marks)

END OF QUESTION PAPER

NB: This paper contains 9 printed pages.